4 Modified Experimental Design

This chapter introduces a revised experimental design that builds on Antinyan et al. (2023) but addresses central limitations of their study. The original results suggest that the impersonal, one-shot setting may have constrained the behavioural impact of communication. The modified design therefore incorporates two features that are characteristic of organisational and interpersonal delegation contexts: repeated interaction and opportunities for direct interpersonal exchange. This design focuses solely on realised behavioural outcomes rather than beliefs. Since actual delegation and return decisions are linked to welfare and payoff consequences, they provide a more economically meaningful basis for assessing how repeated interaction and interpersonal communication shape cooperation and are therefore prioritized in this setup.

4.1 Hypotheses

Experimental evidence shows that repeated trust interactions can sustain higher levels of cooperation because past behaviour becomes informative and reputation can influence future outcomes (Attanasi et al. 2019, pp. 341, 357). In such environments, trustees face stronger incentives to behave reliably, and trustors can update their expectations based on information about past behaviour. This motivates the introduction of repeated rounds in the present design.

Evidence also indicates that richer interpersonal communication such as short face-to-face or dialogic interaction can increase cooperation compared to impersonal, one-sided written communication (Fiedler and Haruvy 2009, p. 720). Even brief interpersonal exchange can reduce social distance and make opportunistic behaviour less attractive. Building on these findings, the modified design examines whether repeated interaction and interpersonal dialogue improve delegation and return behaviour compared to the impersonal one-shot structure of Antinyan et al. (2023). These considerations motivate the following hypotheses:

H1a Delegation rates are higher in repeated interactions than in one-shot settings.
H1b Trustees return higher amounts in repeated interactions than in one-shot settings.
H2a Delegation rates are higher under two-way interpersonal dialogue than under one-sided text-form communication.
H2b Trustees return higher amounts under two-way dialogue than under one-sided text communication.

4.2 Design and Operationalization

To assess the hypotheses in Section 4.1, the modified experiment retains the lost–wallet structure of Antinyan et al. (2023) but incorporates two extensions: repeated interaction and two-way interpersonal communication. The design focuses exclusively on the high-cost condition ($x = 13$), as this is the environment in which the original study observed low delegation rates (35.3%) and large average losses for trustors (€7.1) despite communication. Concentrating on this setting allows for a clearer identification of whether repetition and interpersonal dialogue can improve delegation outcomes.

Participants are assigned either to a one-shot version (R = 1) or to a repeated version consisting of five rounds (R = 5). In repeated treatments, the total number of rounds is not disclosed to participants in order to prevent last-round strategic behaviour. In each round, trustors decide whether to delegate, and trustees choose how much to return. To collect full data, the returns are elicited before the trustees know whether the trustor actually delegated. After every round, trustors evaluate the trustee's behaviour using a binary rating ("trustworthy" / "not trustworthy"). This reputation signal is shown to the next trustor before the communication stage, and participants are randomly re-matched each round so that trustees interact with different partners over time.

Communication varies across two conditions. In the one-way condition, the trustee sends a short, impersonal text message before the trustor chooses whether to delegate, mirroring the original study. In the two-way condition, the pair engages in a brief interpersonal dialogue of three minutes, sufficient to convey a basic personal impression while maintaining experimental control. Combining the repetition factor with the communication protocol produces the 2×2 treatment structure presented in Table 2.

	One-way communication	**Two-way communication**
R = 1	*OW1*	*TW1*
R = 5	*OW5*	*TW5*

Table 2: The four experimental treatments in the modified design

The experiment can be implemented in a standard laboratory environment using individual cubicles or desks with visual dividers to prevent participants from seeing each other's screens. The one-way communication treatment is conducted entirely within the computer interface, where trustees compose a short text message. In the two-way communication treatment, matched pairs briefly meet at a designated communication station separated from other participants by movable partitions or a small enclosed area. The interaction lasts for three minutes to ensure personal impressions while keeping the experiment within a feasible timeframe. After the dialogue, participants return to their cubicles to make their decisions privately.

All decisions, matching procedures, and reputation updates can be administered using a software such as zTree. The software presents the relevant reputation information to trustors before they enter the communication phase and automatically assigns new partners in each round according to the random-rematching protocol. Roles remain fixed throughout the experiment. Participant assignment to treatments should follow randomisation with respect to gender, aiming at approximately balanced gender shares in each cell to reduce the risk that treatment differences are confounded with differences in gender composition. Instructions are displayed stage by stage to avoid information overload and to maintain consistency across rounds. Sessions of approximately 60-90 minutes allow for instructions, communication, five rounds of play, and reputation updates. Participants receive a show-up fee of 3 euros plus the payoff from one randomly selected round. This incentives attentive decision-making while keeping total spendings within a reasonable range.

4.3 Analytical Approach

The analysis is based on the main outcome variables generated by the modified design. Delegation decisions are recorded as a binary variable $c_A \in \{0, 1\}$, where 1 indicates that the trustor delegates. Return behaviour is measured as $c_B \in [0, 20]$, the amount the trustee returns conditional on delegation. After each round, trustees receive a binary reputation signal $r \in \{\text{trustworthy}, \text{not trustworthy}\}$ from the previous trustor. Treatment indicators capture the repetition condition and the communication mode: OW1, OW5, TW1, TW5. These variables form the basis for all behavioural comparisons in the empirical analysis.

To evaluate the hypotheses introduced in Section 4.1, the analysis compares outcomes across treatment pairs that isolate the two mechanisms of interest: repetition and interpersonal dialogue. The effect of repeated interaction (H1a, H1b) can be examined by comparing OW1 with OW5 and, separately, TW1 with TW5. The first comparison identifies whether repetition increases delegation and returns when communication remains impersonal, while the second shows the repetition effect when participants engage in interpersonal dialogue. Considering both contrasts makes it possible to assess whether the influence of repeated interaction is robust across communication environments or variant across the interaction modes.

The effect of interpersonal dialogue (H2a, H2b) can be assessed by comparing OW1 with TW1 and, analogously, OW5 with TW5. The first comparison isolates the communication effect in a pure one-shot setting, whereas the latter examines whether dialogue adds explanatory power in environments with repeated interaction. TW5 measures the combined effect of repetition and personal interaction and allows for result comparisons with OW5 and TW1 to indicate whether dialogue and repetition operate independently, complementarily, or as substitutes in shaping delegation and return behaviour.

Formal inference can be based on regression-based and non-parametric approaches. Logistic or probit models are suitable for analysing the binary delegation variable c_A, as they estimate how treatments affect the probability of delegation. Bounded return behaviour c_B can be analysed using Tobit models, which account for censoring at the limits. In repeated-round treatments, standard errors may be clustered at the participant level to address within-subject correlation. Standard deviations for each treatment can additionally be reported to capture behavioural heterogeneity in return decisions. For comparing outcomes between individual treatments, Wilcoxon-Mann-Whitney tests can be used. These tests do not rely on assumptions about the distribution or functional form of the data, making them well suited for experimental outcomes that may be skewed or heterogeneous.

The design also permits an exploratory analysis of reputational recovery. Because each trustor observes the trustee's most recent reputation signal before making a delegation decision, it becomes possible to investigate whether trustees rated as "not trustworthy" in the previous round still obtain delegation in the subsequent one, and whether this likelihood differs between OW and TW communication modes. Comparing delegation rates for negatively rated trustees across communication conditions yields insight into whether interpersonal dialogue can mitigate reputational penalties or help rebuild trust.

To ensure that effects of plausible magnitude can be detected, a sample-size calculation was carried out using G*Power. Mean differences and standard deviations from the high-cost treatments in Antinyan et al. (2023) served as benchmarks for computing an effect size for a Wilcoxon-Mann-Whitney test. Based on these inputs, a total sample size of approximately 720 participants was determined if every of the four treatments is powered. Further details of the calculation are reported in Appendix C.

4.4 Limitations and Potential Applications

The modified design also comes with several limitations. A first limitation arises from the payment protocol. Since participants must be informed that only one randomly selected round is paid out, they may behave more risk-averse and try to secure acceptable payoffs in each round. This can weaken behavioural mechanisms that typically emerge in repeated trust settings, such as reputation building or other game-theoretic forms of forward-looking behaviour. As a result, the strategic dynamics that repetition is intended to capture may be expressed less strongly than in designs where payoffs accumulate across rounds.

Furthermore, the needed sample size for achieving statistical power of approximately 720 participants is demanding for a laboratory project and would require multiple sessions. This also restricts the inclusion of additional treatments such as a no-communication condition from the base study. Each further treatment would substantially increase the required sample size and reduce feasibility.

A third set of limitations concerns the scope of the design. Focusing exclusively on the high-cost environment ($x = 13$) allows a targeted examination of delegation under demanding conditions but reduces comparability with Antinyan et al. (2023), who study the effect of the opportunity cost of delegation across cost levels. Similarly, the design does not elicit beliefs, which would allow a fuller decomposition of communication effects but would add considerable procedural complexity. As implemented, the study therefore concentrates on behavioural outcomes in the most challenging delegation environment while leaving broader treatment comparisons to future work.

Despite these limitations, the proposed design offers meaningful potential applications. First, it can shed light on how repetition and interpersonal exchange influence delegation decisions in environments where principals possess strong outside options which are settings that mirror organisational practice, project-based teamwork or supervisory roles. Second, the reputational component provides insights into the extent to which trust can be regained after negative signals, which is relevant for internal labour markets, leadership assessment, and organisational learning processes. Third, comparing one-way text communication with brief interpersonal dialogue can reveal how different communication formats shape cooperative behaviour, which is relevant for relational settings in which interactions are often brief and partners have limited prior history. Overall, the experiment has the potential to contribute to a broader understanding of how relational features influence cooperation in high-stakes delegation contexts.